

NILAY SINGH

+91 7982631261 ✦ Gurugram,INDIA

nilay31singh@gmail.com ✦ [LinkedIn](#) ✦ [Portfolio Site](#) ✦ [Github](#)

EDUCATION

B.Tech in Computer Engineering, Thapar University
CGPA: 8.91

Aug 2023 - May 2027

SKILLS

ML & AI	TensorFlow, Scikit-Learn, LangChain, LangGraph, RAG
Backend & Web	Python, FastAPI, React, SQL,HTML,CSS
Data & Storage	Pandas, NumPy, Vector Databases (ChromaDB/FAISS)
Visualization	Matplotlib, Seaborn, Plotly, Streamlit

PROJECTS

KHABRI: AI News Agent

[\[GitHub\]](#)

Autonomous AI agent for real-time news aggregation and summarization.

Tech Stack: Python, FastAPI, React, LangGraph, Tavily Search

- **Engineered** an autonomous agent using **LangGraph** and **Tavily Search** to dynamically aggregate and synthesize news data.
- **Developed** a high-performance REST API backend using **FastAPI** to integrate the agentic workflow with a **React** frontend.

Delhi Air Quality Prediction

[\[GitHub\]](#) [\[Dataset\]](#)

Deep learning time-series forecasting system for environmental monitoring.

Tech Stack: Python, TensorFlow/Keras (GRU, LSTM), Pandas, Scikit-Learn

- **Developed** a time-series forecasting system benchmarking multiple architectures (LSTM, CNN-LSTM); identified **GRU** as the optimal model for capturing extreme pollution spikes.
- **Engineered** 30+ temporal features (lag, rolling, cyclical encoding) across 8 years of data, achieving an R^2 **Score of 0.85** and a low **RMSE of 38.45**.

Mental Health Predictor App

[\[GitHub\]](#) [\[Live Demo\]](#)

Interactive machine learning dashboard for treatment identification analytics.

Tech Stack: Python, CatBoost, Streamlit, Pandas, Plotly

- **Engineered** a robust classification model using **CatBoost** to handle complex categorical industry data, achieving a **Recall of 0.83** and **F1-Score of 0.74**.
- **Deployed** an interactive **Streamlit** UI with **Plotly** visualizations, providing users with a predictive self-reflection quiz and transparent feature importance insights.

RAG-CHATBOT: Local LLM AI

[\[GitHub\]](#)

Privacy-focused NLP semantic search system.

Tech Stack: Python, FastAPI, React, LangChain, Ollama, FAISS

- **Architected** a RAG pipeline utilizing **FAISS** for high-speed vector retrieval across dense document datasets with sub-second latency.
- **Optimized** local inference using **Ollama (Phi-3)**, reducing hallucinations by strictly grounding LLM responses within the retrieved context window.

CERTIFICATIONS

Machine Learning Specialization

[\[View Credential\]](#)

July 2025

Issued by DeepLearning.AI & Stanford University

- **Pioneered** advanced machine learning applications, encompassing **supervised learning** for predictive modeling, **deep neural networks** with **TensorFlow**, and the development of **reinforcement learning** models for complex AI systems.